

Warmup: Deviance

Dengue data

To predict dengue status, you build a model that uses Age, WBC, and PLT as explanatory variables. You fit the model in R, and compute the deviance on the training data:

```
m_dengue1 <- glm(Dengue ~ Age + WBC + PLT,  
                 data = dengue,  
                 family = binomial)  
m_dengue1$deviance
```

```
[1] 5200.823
```

Your friend also builds a model, but they also include HCT in their model:

```
m_dengue2 <- glm(Dengue ~ Age + WBC + PLT + HCT,  
                 data = dengue, family = binomial)  
m_dengue2$deviance
```

```
[1] 5199.946
```

Question

1. Your friend notices that their model has a smaller deviance, and so they argue that their model is preferable to yours. What do you think?